

Mathematical Funditions of Reinforcement Learning and Online Decision Making Homework 1

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Some inequalaites that might be useful:

- Triangle inequality

$$||x + y|| \leq ||x|| + ||y||, \text{ for all } x \in \mathbb{R}^d.$$

- Extended triangle inequality

$$||x + y||^2 \leq 2||x||^2 + 2||y||^2, \text{ for all } x \in \mathbb{R}^d$$

- Power inequality

$$1 + x \leq \exp(x), \text{ for all } x \in \mathbb{R}.$$

Question 1

- Consider an irreducible Markov chain. Prove that if one state has a self-loop (i.e. $P(x, x) > 0$ for some state $x \in \mathcal{X}$) then the Markov chain is aperiodic.
- Provide an example of an irreducible and aperiodic Markov chain that has no self loops.

Question 2 [Extension of Exercise 1.6 in Levin & Peres]

Let P be an irreducible transition matrix of period b .

- Show that the state space $\mathcal{X}$ can be partitioned into b sets $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_b$, where $P(x, y) > 0$ only if $x \in \mathcal{C}_i$ and $y \in \mathcal{C}_{i+1}$ for some i (where the addition $i+1$ is modulo b).

- b) Suppose that $|\mathcal{X}| = n$ and let $n_i = |\mathcal{C}_i|$ and

$$N_i = \sum_{j=1}^i n_j$$

for $i = 1, 2, \dots, b$. Let us now enumerate the state such that $x_1, \dots, x_{N_1} \in \mathcal{C}_1$ and $x_{N_{i-1}+1}, \dots, x_{N_i} \in \mathcal{C}_i$ for $i = 2, \dots, b$.¹ Characterize the sparsity structure of the associated transition matrix.

Question 3 [Extension of Exercise 1.7 in Levin & Peres]

A transition matrix P is symmetric if $P(x, y) = P(y, x)$ for all x and y in $\mathcal{X}$.

- a) Show that if P is symmetric then the uniform distribution on $\mathcal{X}$ is a stationary distribution of P .
- b) Does there exist a non-symmetric transition matrix with the uniform distribution as its stationary distribution. Prove or disprove.

Question 4

Consider a irreducible and aperiodic Markov chain with a state space $\mathcal{X}$ and a transition matrix P . We have a reward function $R : \mathcal{X} \rightarrow \mathbb{R}_+$, where $R(x)$ represents the reward for being in state $x \in \mathcal{X}$. We assume that $R(x)$ is bounded and $|R(x)| \leq R_{\max}$ for all $x \in \mathcal{X}$.

- a) Consider the average reward value function

$$V_T(x_0) = \frac{1}{T+1} \mathbf{E} \left[\sum_{t=0}^T R(X_t) | X_0 = x_0 \right]$$

Prove that

$$V(x_0) := \lim_{T \rightarrow \infty} V_T(x_0) = \sum_{x \in \mathcal{X}} \pi(x) R(x)$$

where $\pi : \mathcal{X} \rightarrow [0, 1]$ is the unique stationary distribution of the Markov chain.

- b) Suppose that the mixing time $\tau_{\text{mix}}(\epsilon)$ of the Markov chain is known. Consider the following algorithm to estimate $V(x_0)$:

- Run N simulations starting at $x_0 \in \mathcal{X}$ with length $\tau_{\text{mix}}(\epsilon)$ where

$$X_n = (X_{n,0}, X_{n,1}, \dots, X_{n,\tau_{\text{mix}}(\epsilon)})$$

is the realization of the n -th simulation.

¹Note that for $i = 2$ the series $n_1 + \dots + n_{i-1} = \sum_{j=1}^{i-1} n_j = n_1$.

- Set $Y_n(\epsilon) = X_{n, \tau_{\text{mix}}(\epsilon)}$ and compute the estimator:

$$\hat{V}_N = \frac{1}{N} \sum_{n=1}^N R(Y_n(\epsilon))$$

Prove that the estimator has the following bound on the MSE:

$$\mathbf{E}(\hat{V}_N - V)^2 \leq R_{\max}^2 \left(\frac{1}{N} + 4\epsilon^2 \right)$$

Hint: The Popoviciu's inequality might be useful: https://en.wikipedia.org/wiki/Popoviciu%27s_inequality_on_variances

- c) Use the result in b) to show that for any $\epsilon > 0$ we find an ϵ accurate estimator with respect to the MSE (i.e., such that $(\hat{V}_N - V)^2 \leq \epsilon$) by taking

$$\left\lceil \frac{R_{\max}^2}{\epsilon} \right\rceil \left\lceil \frac{1}{2} \log \left(\frac{8R_{\max}^2}{\epsilon} \right) \right\rceil \tau_{\text{mix}} \left(\frac{1}{4} \right) \text{ samples.}$$

Hint: You can use the following mixing time bound in Eq. (4.34) in Levin & Peres:

$$\tau_{\text{mix}}(\epsilon) \leq \lceil \log_2 \epsilon^{-1} \rceil \tau_{\text{mix}}(1/4).$$

Question 5

Let $T : \mathcal{X} \rightarrow \mathcal{X}$ be an α -contraction in some norm $\|\cdot\|$.

- Prove that if T has a fixed point $x^* \in \mathcal{X}$ then it is unique.
- Give an example of a α -contraction $T(\cdot)$ and $\mathcal{X}$ with no fixed point.
- The set $\mathcal{X}$ together with the metric $d(x, y) = \|x - y\|$ is called a complete metric space if every Cauchy sequence in $\mathcal{X}$ has a limit in $\mathcal{X}$.² Prove if $\mathcal{X}$ is a complete metric space, then $T(\cdot)$ has a fixed point in $\mathcal{X}$.

Question 6

Consider an undirected graph $(\mathcal{N}, \mathcal{E})$ with nodes $\mathcal{N}$ and edges $\mathcal{E}$. In this graph each edge represents a communication link, so that two nodes $n, m \in \mathcal{N}$ can communicate if and only if $(n, m) \in \mathcal{E}$. We assume throughout that the graph is connected, i.e., there exists a path between every two nodes $n, m \in \mathcal{N}$.

²A sequence $x_1, x_2, \dots$ of elements in $\mathcal{X}$ is called a Cauchy sequence if for every $\epsilon > 0$ there exists N such that $\|x_n - x_m\| \leq \epsilon$ for all $n, m \geq N$. An example of a metric space that is not complete is the set of rational numbers $\mathbb{Q}$ since, e.g., $\sqrt{2}$ is not rational, but we can create a sequence of rational number that converge to $\sqrt{2}$.

Suppose that each node has a scalar value $x_n \in \mathbb{R}$ and the nodes want to find a local estimate of the global average

$$\bar{x} = \frac{1}{N} \sum_{n=1}^N x_n$$

where $N = |\mathcal{N}|$. There is a simple distributed algorithm for estimating $\bar{x}$ where each node computes local averaging based on information from its communication neighbors. In particular, let $x_n(k)$ be node's n estimation of $\bar{x}$ at iteration k , where $x_n(0) = x_n$. Then at each iteration $k \geq 0$ node n computes a weighed average

$$x_n(k+1) = \sum_{m=1}^N W_{nm} x_m(k)$$

where $W \in \mathbb{R}^{N \times N}$ is a double stochastic matrix that inherits the sparsity structure of the communication graph $(\mathcal{N}, \mathcal{E})$. In particular, each row and each column of W sums to one, i.e.,

$$W\mathbf{1} = \mathbf{1} \quad \text{and} \quad \mathbf{1}^T W = \mathbf{1}^T$$

where $\mathbf{1} \in \mathbb{R}^N$ is a column vector of ones. Moreover, $W_{nn} > 0$ and $W_{nm} > 0$ if there is an edge between node n and m , i.e., $(n, m) \in \mathcal{E}$. Otherwise, $W_{nm} = 0$. Due to this sparsity, the algorithm can be implemented based on using only local communications, to perform one iteration each node only needs information from its neighbors.

The algorithm can be written more compactly as

$$x(k+1) = Wx(k)$$

where $x(k) = [(x_1(k), \dots, x_N(k))]^T$. To goal of this exercise is to prove that the iterates $x(k)$ converge to the average vector $\bar{x}\mathbf{1}$ at a linear rate.

a) Show that the algorithm

$$x(k+1) = Wx(k)$$

is not contractive in any norm.

b) Show that the set of fixed points of the algorithms is

$$\text{Span}(\mathbf{1}) = \{a\mathbf{1} \in \mathbb{R}^N | a \in \mathbb{R}\}.$$

Hint: The Perron–Frobenius theorem might be helpful. One result of the theorem states that if A is an irreducible non-negative matrices with spectral radius $r = \rho(A)$ then r is a simple eigenvalue of A and both right and left eigenspaces associated with r are one-dimensional. You may use the fact that the spectral radius of W is 1 and W is irreducible matrix.

c) Consider the set

$$\mathcal{A}(\bar{x}) = \left\{ x \in \mathbb{R}^N \mid \frac{1}{N} \sum_{n=1}^N x_n = \bar{x} \right\}$$

show that if $x \in \mathcal{A}(\bar{x})$ then $Wx \in \mathcal{A}(\bar{x})$.

d) Show that if $x \in \mathcal{A}(\bar{x})$ then

$$\|Wx - \bar{x}\mathbf{1}\|_2 \leq \sigma \|x - \bar{x}\mathbf{1}\|_2,$$

where

$$\sigma = \|W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\|_2 < 1.$$

Conclude that W is a pseudo-contraction on the set $\mathcal{A}(\bar{x})$.

e) Another approach to study the convergence of this algorithm is to consider the eigenvalues of W . Show that

$$\|x(k) - \bar{x}\mathbf{1}\| \leq A|\lambda_2|^k \quad \text{for all } k$$

where A is some constant and λ_2 is the second largest eigenvalue of W in absolute value. You may assume that W is symmetric matrix.

Hint: Again, the Perron–Frobenius theorem might be helpful. You may use the fact that the spectral radius of W is 1.

Question 7

Consider the following gradient descent algorithm

$$x(k+1) = x(k) - \gamma(\nabla f(x(k)) + e(k))$$

where $e(k)$ are IID noise zero mean noise with variance σ^2 . We make the following assumptions.

- $f(\cdot)$ is L -smooth, i.e., following holds for all x and y

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|.$$

- $f(\cdot)$ is μ -convex, following holds for all x and y

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|x - y\|^2.$$

Show that if $\gamma \in (0, 2/(L + \mu))$ then

$$\mathbf{E}\|x(k+1) - x^*\|^2 \leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right)^{k+1} \mathbf{E}\|x(0) - x^*\|^2 + \gamma\sigma^2 \frac{\mu + L}{2\mu L}.$$

Conclude that even though σ^2 , we can obtain arbitrarily good solution accuracy by proper tuning of γ . What is the cost of high solution accuracy?

Hint: You may use the following inequality without proof. If $f(\cdot)$ is L -smooth and μ -convex then

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{L\mu}{L + \mu}\|x - y\|^2 + \frac{1}{\mu + L}\|\nabla f(x) - \nabla f(y)\|^2 \quad \text{for all } x \text{ and } y.$$

Question 8

Consider a sequence of random variable $X_1, X_2, X_3, \dots$. Often we would like to understand if the sequence converges to some limit. For random variables there are many reasonable definition of convergence that are not equivalent. Two popular definitions are as follows.

- **Almost sure convergence:** We say that a X_k converges almost surely to X if

$$\mathbf{P} \left(\lim_{k \rightarrow \infty} X_k = X \right) = 1$$

- **Convergence in probability:** We say that a X_k converges in probability if for all $\epsilon > 0$ the following holds:

$$\lim_{k \rightarrow \infty} \mathbf{P} (|X_k - X| \geq \epsilon) = 0$$

It can be proved that if X_k converges almost surely then X_k converges in probability (you don't need to prove that). However, the converse does not hold, it is possible to find a sequence X_k that converges in probability but not almost surely. Provide example of such a sequence X_k .

Question 9

Two useful inequalities in probability theory are Markov's inequality and Chebychev's inequality:

- **Markov's inequality:** If X is a non-negative random variable, then for all α

$$P(X \geq \alpha) \leq \frac{\mathbf{E}X}{\alpha}$$

- **Chebychev's inequality:** Let Y be an arbitrary random variable with finite mean μ . Then for all $\alpha > 0$

$$P(|X - \mu| \geq \alpha) \leq \frac{\mathbf{Var}X}{\alpha^2}$$

- a) Use Markov's inequality to prove Chebychev's inequality.
- b) Use Chebychev's inequality to prove the weak law of large numbers. In particular, let $X_1, X_2, \dots, X_n$ be a sequence of IID random variables, each having the mean μ and variance σ^2 . Consider the random variable

$$S_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Prove that

$$\mathbf{P} (|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2 n}.$$

Conclude that the random variable S_n converges in probability to μ .

Side-note: This is called the weak law of large numbers because the convergence is in probability. It is possible to establish almost sure convergence under more small additional assumptions, this is the strong law of large numbers.

- c) Let $\epsilon > 0$ and $\delta \in (0, 1)$ be given. How many samples do you need to collect to ensure that S_n is an ϵ -approximation of μ with probability at least $1 - \delta$. In other words, how should we choose n so that

$$|S_n - \mu| \leq \epsilon$$

with probability at least $1 - \delta$.